

Jesús Emilio Domínguez Russell

Email: jedguezr@umd.edu

Website: jedguez.github.io

EDUCATION

University of Maryland, College Park

Ph.D. in Mathematics – Advisor: Amin Gholampour.

Universidad Autónoma de Sinaloa

B.Sc. in Mathematics.

College Park, MD, USA

August 2021–Present

Culiacán, Sinaloa, México

August 2016–July 2021

PUBLICATIONS/PREPRINTS

- [1] E. Dominguez and A. Gholampour, *Virtual cycles of 3-term complexes and the hilbert schemes of surfaces*, 2026. arXiv: 2606.29542 [math.AG].
- [2] O. Arizmendi and E. Dominguez, “Barabasi–albert trees are hypoenergetic”, *Boletín de la Sociedad Matemática Mexicana*, vol. 28, no. 3, p. 72, Sep. 2022, ISSN: 2296-4495.
- [3] J. E. Domínguez and C. Segovia, “Extending free actions of finite groups on surfaces”, *Topology and its Applications*, vol. 305, p. 107 898, 2022, ISSN: 0166-8641.

TEACHING

- **Instructor for the University of Maryland** Summer 2023, Summer 2024
Calculus I (MATH140)
- **Teaching Assistant for the University of Maryland** Spring 2021–Fall 2026
Introduction to Linear Algebra (MATH240), Calculus I (MATH140), Elementary Calculus I (MATH120), Precalculus (MATH115)
- **Grader for the University of Maryland** Fall 2024, Summer 2026
Advanced Calculus II (MATH411), Applications of Linear Algebra (MATH401).

TALKS

- **Relative GW theory** March 2026
RIT on Algebraic Geometry – Logarithmic Geometry, at University of Maryland
- **Virtual Classes and GW theory** February 2026
RIT on Algebraic Geometry – Logarithmic Geometry, at University of Maryland
- **Moduli of vector bundles and Brill-Noether Theory** February 2025
RIT on Algebraic Geometry – K3 Surfaces, at University of Maryland
- **Intersections in Algebraic Geometry** May 2024
Student Seminar, at Universidad Autónoma de Sinaloa, Mexico
- **The Inverse Galois Problem** April 2024
2024 Seymour Goldberg and Monroe Martin Spotlight Talks, at University of Maryland

MENTORING

- **Counselor at Brin Maryland Mathematics Camp** Summer 2024, Summer 2025, Summer 2026
Courses in Knot Theory and Mathematical Biology
- **Mentor for Undergraduate Directed Reading Program at UMD** Spring 2023, Fall 2025
Topics: “Modular forms and their applications”, “Space-filling curves”
- **Coach for Sinaloa state team in the Mexican Math Olympiads** 2016–2020

AWARDS

- **Herbert A. Hauptman Fellow** 2025, 2026
- **Dean’s Fellowship, University of Maryland** 2021–2023